

Controllable Nondegenerate p-Type Doping of Tungsten Diselenide by Octadecyltrichlorosilane

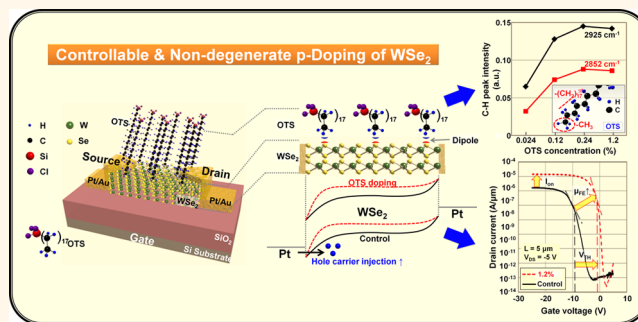
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ABSTRACT Despite heightened interest in 2D transition-metal dichalcogenide (TMD) doping methods for future layered semiconductor devices, most doping research is currently limited to molybdenum disulfide (MoS₂), which is generally used for n-channel 2D transistors. In addition, previously reported TMD doping techniques result in only high-level doping concentrations (degenerate) in which TMD materials behave as near-metallic layers. Here, we demonstrate a controllable nondegenerate p-type doping (p-doping) technique on tungsten diselenide (WSe₂) for p-channel 2D transistors by adjusting the concentration of octadecyltrichlorosilane (OTS).

This p-doping phenomenon originates from the methyl (–CH₃) functional groups in OTS, which exhibit a positive pole and consequently reduce the electron carrier density in WSe₂. The controlled p-doping levels are between 2.1×10^{11} and 5.2×10^{11} cm^{–2} in the nondegenerate regime, where the performance parameters of WSe₂-based electronic and optoelectronic devices can be properly designed or optimized (threshold voltage[†], on-/off-currents[†], field-effect mobility[†], photoresponsivity[†], and detectivity[†] as the doping level increases). The p-doping effect provided by OTS is sustained in ambient air for a long time showing small changes in the device performance (18–34% loss of ΔV_{TH} initially achieved by OTS doping for 60 h). Furthermore, performance degradation is almost completely recovered by additional thermal annealing at 120 °C. Through Raman spectroscopy and electrical/optical measurements, we have also confirmed that the OTS doping phenomenon is independent of the thickness of the WSe₂ films. We expect that our controllable p-doping method will make it possible to successfully integrate future layered semiconductor devices.



KEYWORDS: OTS · WSe₂ · nondegenerate doping · electronic device · optoelectronic device

Two-dimensional (2D) transition-metal dichalcogenides (TMDs) with layered structures are considered promising materials for next-generation wearable, flexible, stretchable, and transparent electronics because of their superior electrical, optical, and mechanical properties.^{1–13} Because their thickness is scalable down to a monolayer and they exhibit a van der Waals epitaxial structure without surface dangling bonds, TMD-based thin film transistors (TFTs) are free of the short channel effect (SCE) and carrier mobility degradation caused by surface oxidation/scattering.^{1,2} TMDs are also expected to be popular in optoelectronic applications for wide-spectral photodetectors and light emitting

diodes (LEDs) because their energy band-gap varies with layer thickness.^{5–12} However, the lack of a reliable and controllable doping method, which is essential for inducing a Fermi level shift that subsequently enables modulations of electrical and optical properties, currently prevents such TMD-based electronic and optoelectronic devices from being successfully integrated.

Although much work is currently being done in the field of doping in TMDs, most of it is limited to molybdenum disulfide (MoS₂),^{15–20} which forms a low electron barrier contact with metals and is generally used for n-channel transistors. As a result, more research is needed on the doping of tungsten diselenide (WSe₂), which shows a

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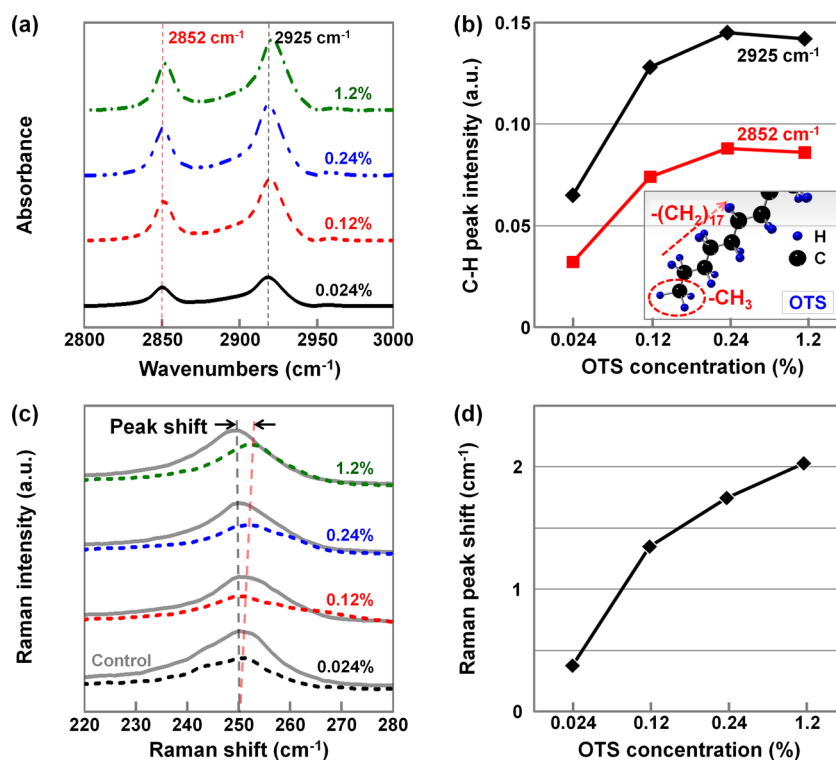


Figure 1. (a) FT-IR spectra of WSe₂ films doped by OTS at different concentrations (0.024%, 0.12%, 0.24%, and 1.2%) and (b) C–H peak intensity data as a function of OTS concentration extracted from the peaks at 2852 and 2925 cm^{−1}. Inset: schematic diagram of OTS with –CH₃ and –(CH₂)₁₇ groups. (c) Raman spectra of the OTS-doped WSe₂ films and (d) Raman peak shift data as a function of OTS concentration extracted from the peaks at 250 cm^{−1}.

relatively low hole barrier contact with metals and can therefore be used to integrate p-channel transistors to realize 2D TMD-based applications. Fang *et al.* reported an NO₂ molecule-based p-doping¹⁴ and potassium-based n-doping process¹⁵ on WSe₂ using the surface charge-transfer mechanism. They used it to form highly p-/n-doped source/drain (S/D) regions in p-/n-channel TMD-based TFTs. They also controlled the n-doping level by adjusting the exposure time to potassium, but both the p-/n-doping techniques resulted in high-level doping concentrations (degenerate) in which WSe₂ behaved as a near-metallic layer. Therefore, doping controllability in a light doping regime (nondegenerate), in which WSe₂ serves as a semiconducting material, is a critical point in the design and fabrication of WSe₂-based electronic and optoelectronic devices. However, it is difficult to achieve nondegenerate doping on 2D semiconductors (including TMDs) in a controllable fashion because the precisely controllable ion implantation technique cannot be applied to the doping because of crystal damage that occurs while implanting. Here, we demonstrate a controllable nondegenerate p-doping technique on trilayer, few layer, and bulk WSe₂ films by octadecyltrichlorosilane (OTS). Methyl (–CH₃) functional groups in OTS have a positive pole and consequently reduce the electron carrier density in WSe₂, showing a p-doping phenomenon. We have systematically investigated the proposed p-doping method in terms of controllability and

air-stability through Fourier transform infrared spectroscopy (FT-IR), Raman spectroscopy, capacitance–voltage (C–V) measurement, and electrical/optical measurements (*I*_D–*V*_G and *I*_D–*V*_D with/without exposure to a 785 nm wavelength laser). By adjusting the concentration of OTS in hexane, we control the p-doping level of WSe₂, which can affect the performance (threshold voltage, on-/off-currents, mobility, photoresponsivity, and detectivity) of WSe₂-based electronic and optoelectronic devices. The p-doping effect by OTS is sustained in ambient air for a long time with only small changes in the device performance. Furthermore, performance can also be almost completely recovered by a thermal annealing process.

RESULTS AND DISCUSSION

Formation and Doping Control of OTS on WSe₂ Films. First, we performed FT-IR and Raman spectroscopy analyses on OTS/WSe₂/SiO₂ samples to investigate, respectively, (1) the controllability of OTS concentration in hexanes and (2) the doping effect of OTS on WSe₂. Figure 1 shows (a) the FT-IR spectra (between 2800 and 3000 cm^{−1}) measured on the OTS/WSe₂/SiO₂ samples with different OTS concentrations (0.024%, 0.12%, 0.24%, and 1.2%) and (b) the extracted C–H peak intensity as a function of OTS concentration. We found that the peaks for the –CH₂ groups in a long alkyl group chain (–(CH₂)₁₇) of OTS are at 2852 and 2925 cm^{−1},²¹ and their peak intensity increased as

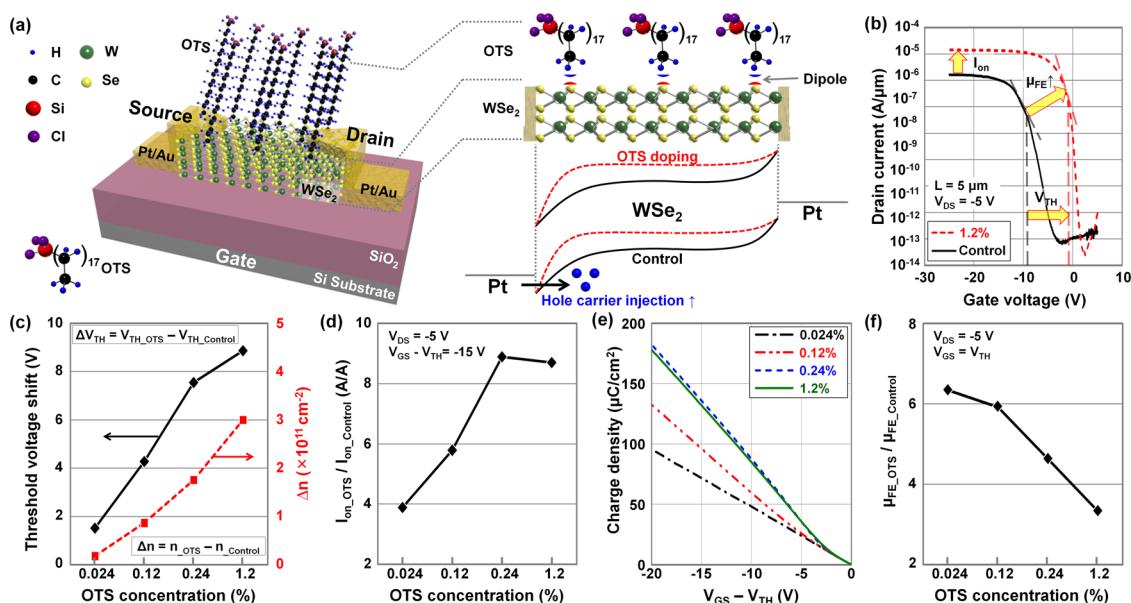


Figure 2. (a) Schematic diagrams of back-gated transistor fabricated on OTS-doped WSe₂ and the energy band diagrams of Pt–WSe₂–Pt junctions. (b) I_D – V_G characteristics of the transistors fabricated on (black) undoped and (red) 1.2% OTS-doped WSe₂ films ($L = 5 \mu\text{m}$ and $V_{DS} = -5 \text{ V}$). (c) Threshold voltage shift ($\Delta V_{TH} = V_{TH,OTS} - V_{TH,Control}$) and carrier concentration increase ($\Delta n = n_{OTS} - n_{Control}$) as a function of OTS concentration (0.024%, 0.12%, 0.24%, and 1.2%). (d) On-current ratio (I_{on} ratio = $I_{on,OTS}/I_{on,Control}$) at $V_{GS} - V_{TH} = -15 \text{ V}$ and $V_{DS} = -5 \text{ V}$ as a function of OTS concentration. (e) Charge density vs $V_{GS} - V_{TH}$ measured on Pt-OTS-doped WSe₂–SiO₂–Si back gate structure. (f) Field-effect mobility ratio (μ_{FE} ratio = $\mu_{FE,OTS}/\mu_{FE,Control}$) at $V_{GS} = V_{TH}$ and $V_{DS} = -5 \text{ V}$ as a function of OTS concentration.

the OTS concentration rose from 0.024% to 0.24%. Although the OTS concentration was successfully controlled below 0.24%, the peak intensity was saturated above 0.24%, indicating that a high OTS concentration was already achieved in the 0.24% sample. Parts c and d, respectively, of Figure 1 present the Raman spectra measured on the OTS/WSe₂ samples and their peak shift values extracted after performing a 120 °C anneal for the formation of OTS layers. Because both the E'_{2g} and A'_g modes for WSe₂ are close to 250 cm⁻¹, we found only a single peak at $\sim 250 \text{ cm}^{-1}$.⁷ The blue-shift of the single peak after OTS formation seems to be related to the p-doping phenomenon, based on the Raman peak changes previously reported on doped MoS₂.^{19,22} However, Javey *et al.* and Chen *et al.* recently reported inconsistent Raman peak shift results, observing no change after p-doping of WSe₂ by NO_x²³ and a red-shift after Au nanoparticle-based p-doping,²⁴ respectively. Further research related to Raman analysis of p-doped WSe₂ films will be needed to resolve this issue. As shown in Figure 1d, this blue-shift value continuously increased from 0.372 to 2.028 cm⁻¹ as the OTS concentration increased from 0.024% to 1.2%, probably indicating that a higher p-doping level on WSe₂ is achieved by a more highly concentrated OTS layer. Although the Raman peak shift was observed in the WSe₂ samples treated by various solvents used in this work ($\Delta = 0.08 \text{ cm}^{-1}$ for DI water, 0.08 cm⁻¹ for acetone, 0.12 cm⁻¹ for IPA, 0.17 cm⁻¹ for hexane, and 0.01 cm⁻¹ for toluene), it was negligibly small compared with that of OTS-doped samples (0.372–2.028 cm⁻¹).

Electrical Characterization of OTS-Doped WSe₂ Electronic Devices. Figure 2a shows a schematic diagram of a WSe₂ transistor doped by an OTS layer along with the energy band diagram of Pt-undoped/doped WSe₂–Pt junctions with a negative V_{DS} . To reconfirm the p-doping phenomenon on WSe₂ shown in the previous Raman spectroscopy analysis, I_D – V_G and C – V characteristics of the doped WSe₂ transistors were investigated with OTS concentrations between 0.024% and 1.2%. The lateral electric field (E_{Lateral}) between the source and drain electrodes was fixed at $-1 \text{ V}/\mu\text{m}$ ($V_{DS}/L = -5 \text{ V}/5 \mu\text{m}$). Compared to the control sample (undoped WSe₂), electron carriers further accumulated at the interface between OTS and WSe₂ because the CH₃ group in OTS has positive pole,²⁰ subsequently making the WSe₂ region p-doped. This p-doping phenomenon on WSe₂ by the formation of a molecular dipole eventually influenced the tunneling of hole carriers from the source to the WSe₂. As shown in Figure 2a, the WSe₂ energy band is shifted up by the p-doping, and it increases the electric field at the source–WSe₂ junction, thereby causing the Schottky barrier lowering effect, enhancing the tunneling of holes and reducing the contact resistance. This effect becomes severe when a higher electric field is applied to a metal–semiconductor junction, according to $\Phi_{H,\text{eff}} = \Phi_H - \Delta\Phi_H$, where $\Delta\Phi_H = [(qE)/(4\pi\epsilon)]^{1/2}$. However, this effect is not expected to occur in the other drain–WSe₂ junction because the bend shift by p-doping decreases the electric field there. As a result, a positive shift in threshold voltage (V_{TH}) from -9.1 V to

-0.45 V ($\Delta V_{\text{TH}} = +8.65$ V) was observed after WSe₂ p-doping using a 1.2% OTS layer. In addition, the on-current was increased by about 10 times from 1.64×10^{-6} A/ μm to 1.42×10^{-5} A/ μm , also improving the field-effect mobility (μ_{FE}). This mobility improvement is thought to originate from an inaccurate mobility extraction method that does not exclude the effect of contact resistance.¹ We then confirmed the possibility of controlling ΔV_{TH} by adjusting OTS concentration, as shown in Figure 2c. A further positive shift in V_{TH} was observed as the OTS concentration increased from 0.024% to 1.2% because a more concentrated OTS layer with more positive charges attracted more electrons to the OTS–WSe₂ interface, subsequently increasing the p-doping level in WSe₂. The 2D sheet doping concentration (n) extracted from the relation $n = I_{\text{D}}L/qW\mu V_{\text{DS}}$ is also consistent with the rising trend in V_{TH} as a function of OTS concentration, showing a significant increase in Δn ($= n_{\text{OTS}} - n_{\text{Control}}$) from 1.6×10^{10} to 3.0×10^{11} cm⁻². Based on previously reported concentration data ($\sim 2.5 \times 10^{12}$ cm⁻² for potassium-doped WSe₂ and $\sim 2.2 \times 10^{12}$ cm⁻² for NO₂-doped WSe₂),^{14,15} our p-doping concentration level (between 2.1×10^{11} and 5.2×10^{11} cm⁻²) seems to be in the nondegenerate regime. In Figure 2d, which shows the on-current ratio ($= I_{\text{onOTS}}/I_{\text{onControl}}$) as a function of OTS concentration, we also observed that an increase degree in on-current after OTS doping becomes stronger as the OTS concentration increases. Thus, more hole carriers might accumulate at the WSe₂–SiO₂ interface at the same V_{GS} ($= V_{\text{TH}} - 15$ V) bias because of reduced effective hole barrier height for carrier injection (from source to WSe₂) through OTS doping. To support that claim, we measured the C–V characteristics on the OTS-doped WSe₂ samples and plotted the extracted charge density (Q) from the C–V curves as a function of $V_{\text{GS}} - V_{\text{TH}}$ in Figure 2e. A more detailed explanation of the charge density extraction from C–V characteristics is in Supporting Information (Figure S2). As the V_{GS} negatively increases beyond V_{TH} , the hole charge density increases because hole carriers begin coming to the WSe₂–SiO₂ interface through WSe₂ from the metal contact. In particular, the amount of accumulated hole carriers increases from 71.7 to 132 $\mu\text{C}/\text{cm}^2$ at $V_{\text{GS}} - V_{\text{TH}} = -15$ V as the OTS concentration increases to 1.2%. Here, the contact resistance (R_{Contact}) in the Pt–WSe₂ junction seems to cause a voltage drop (V_{Contact}) that reduces the effective voltage ($= V_{\text{Applied}} - V_{\text{Contact}}$) applied to the SiO₂ region. When OTS doping concentration increases, we expect the contact resistance to be reduced because of the decrease in the effective hole barrier height. Because of this contact resistance reduction effect in highly OTS-doped devices, at a negative V_{GS} lower than V_{TH} , a higher effective voltage seems to be applied on the SiO₂, consequently increasing its charge density. However, the hole carrier density was saturated beyond 0.24%

OTS, which coincides with the saturation phenomenon in the on-current ratio. As shown in Figure 2f and the Supporting Information (Figure S1b), we also extracted field-effect mobility data at $V_{\text{DS}} = -5$ V and $V_{\text{GS}} = V_{\text{TH}}$ and observed that the initial values (30 ± 4 cm² V⁻¹ s⁻¹) of the control devices increased up to ~ 192 cm² V⁻¹ s⁻¹ in the 0.024% OTS-doped sample (~ 105 cm² V⁻¹ s⁻¹ in the 1.2% OTS-doped sample). Field-effect mobility of 110–250 cm²/(V s) was previously reported in WSe₂-based TFTs.^{2,14,15} The mobility values (105–192 cm²/(V s) in 0.024–1.2%) extracted at $V_{\text{GS}} = V_{\text{TH}}$ in this work, which we used for the calculation of carrier concentration, are comparable to the reported values. In particular, the maximum value obtained at different V_{GS} ($= V_{\text{TH}} - 1$ V) is ~ 250 cm²/(V s) in the case of a 0.024% OTS-doped device, which is the same as the highest mobility value previously reported. We reconfirmed the p-type doping controllability of WSe₂ by adjusting the OTS concentration based on the observation that the degree of increase in the field-effect hole mobility decreased as a function of OTS concentration. The reduction in field-effect hole mobility can be attributed to the increased scattering probability at the channel region caused by a higher accumulated hole concentration.

Air-Stability Analysis of OTS-Doped Electronic WSe₂ Devices.

We then monitored the extracted V_{TH} values on the OTS-doped WSe₂ transistors according to air-exposure time to evaluate the air-stability of the doping. Figure 3a shows the V_{TH} shift values as a function of air-exposure time in WSe₂ transistors with different p-doping levels. The V_{TH} values positively shifted by OTS doping move in the negative direction for the first 60 h of air-exposure time, indicating that the degree of p-doping is relieved during air-exposure. However, we found a small V_{TH} shift phenomenon beyond 60 h of air-exposure compared to what we observed for the first 60 h. In the case of highly doped WSe₂ transistors coated with 0.24% and 1.2% OTS layers, we observed relatively high ΔV_{TH} values (1.42 V for 0.24% device and 1.92 V for 1.2% device) because many OTS molecules are affected by moisture in air. Although the p-doping effect is partially relieved during the first 60 h of air-exposure (18–34% loss of ΔV_{TH} initially achieved by OTS doping), it becomes stable in air after 60 h, and also the ΔV_{TH} loss is recovered again by an additional thermal anneal at 120 °C. In addition, we observed a very small reduction in the on-currents extracted at $V_{\text{GS}} - V_{\text{TH}} = -15$ V after air exposure for 120 h. In the device doped by 1.2% OTS, the on-current was reduced by only $\sim 31\%$ (from ~ 68 μA to ~ 47 μA) after 36 h of air-exposure and by $\sim 60\%$ (from ~ 68 μA to ~ 27 μA) after 120 h. In the highly p-doped WSe₂ TFT by NO₂ reported by Peida *et al.*, the on-current decreased from ~ 22 to ~ 6 μA ($\sim 72\%$ reduction) after 36 h of air exposure.²³ The on-current monitoring data for the other OTS-doped samples (0.024%, 0.12%, and 0.24%)

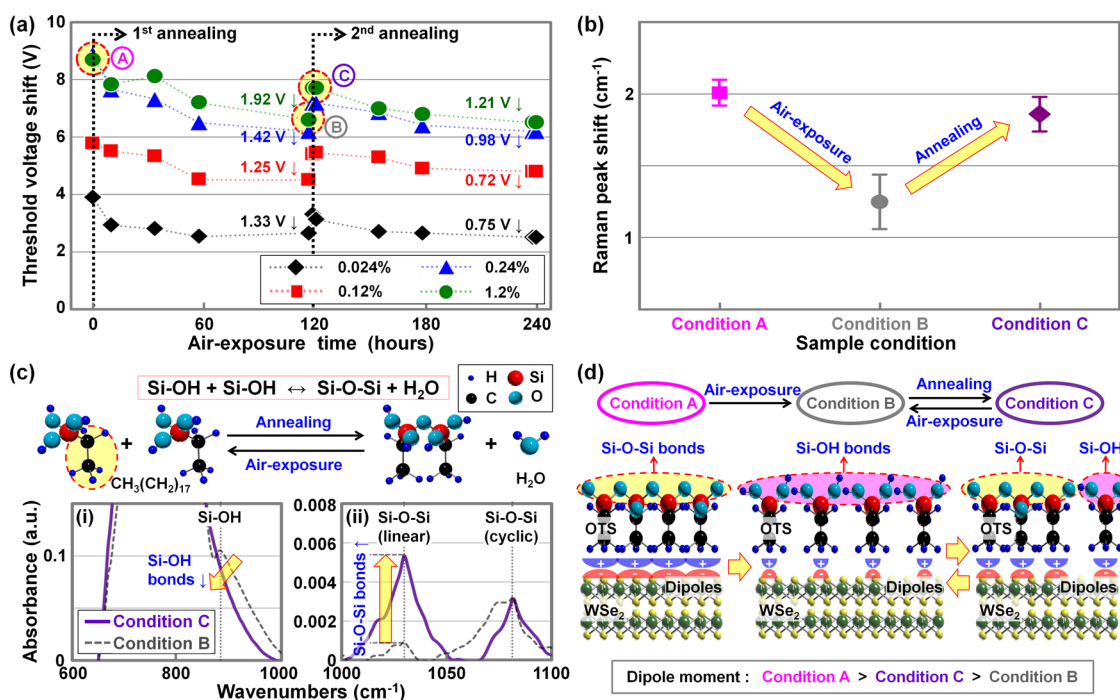


Figure 3. (a) Threshold voltage shift ($\Delta V_{TH} = V_{TH-OTS} - V_{TH-Control}$) of OTS-doped WSe₂ transistors as a function of air-exposure time. Here, after exposure to air for 120 h, the devices were annealed again at 120 °C. Condition A: right after OTS doping and annealing. Condition B: after air exposure for 120 h. Condition C: after additional 120 °C anneal. (b) Raman peak shift values of the three conditions, which were extracted from the peaks at 250 cm^{-1} . (c) Schematic diagram showing the possible chemical reaction between conditions B and C, and FT-IR spectra of OTS under conditions B and C (i) Si–OH peak at $\sim 880 \text{ cm}^{-1}$ and (ii) Si–O–Si peak at $\sim 1030 \text{ cm}^{-1}$. (d) Schematic diagrams predicting how chemical changes in OTS after long periods of air exposure (condition B) and 120 °C recovery annealing (condition C) affect the dipole moments at the WSe₂–OTS interface.

can be found in Figure S4 (Supporting Information). Raman analysis performed on the WSe₂ samples under the three conditions also coincides with the V_{TH} and on-current monitoring data (condition A, right after OTS doping and annealing; condition B, after air exposure for 120 h; condition C, after air exposure for 120 h + additional 120 °C anneal). As shown in Figure 3b, Raman peak shift values extracted from the peaks at 250 cm^{-1} were reduced from 2.01 to 1.25 cm^{-1} after 120 h of air exposure, indicating a weakened p-doping phenomenon. However, the peak shift value was almost recovered (up to 1.86 cm^{-1} from 1.25 cm^{-1}) after an additional 120 °C anneal, as already confirmed in the previous V_{TH} and on-current monitoring experiment. Because the methyl (–CH₃) and polar head (–SiCl₃) functional groups in OTS have positive and negative poles, respectively, due to the asymmetric structure of OTS, the electron carrier density in WSe₂ in contact with the –CH₃ group is reduced, and consequently, the p-doping effect is apparent. The SiCl₃ almost immediately reacts with moisture and forms Si(OH)₃ when OTS is exposed in air. The –OH group attached to the Si atom is expected to react with another –OH group right after the 120 °C anneal. Consequently, it seems to form Si–O–Si bonds between OTS molecules (during this reaction, H₂O is also produced). After exposure of the devices to air for a long time, some of the Si–O–Si bonds are expected to

break through reaction with H₂O, subsequently increasing Si–OH bonds. As shown in Figure 3c, we observed the FT-IR peak related with Si–OH bonds ($\sim 880 \text{ cm}^{-1}$) in the WSe₂ sample exposed to air for 120 h. However, this Si–OH bond disappeared after the 120 °C anneal and the peak for Si–O–Si bonds ($\sim 1030 \text{ cm}^{-1}$) increased, indicating that Si–O–Si bonds were formed again through the reaction between –OH groups. Therefore, as shown in Figure 3d, compared to the case of Si–O–Si bonding, the Si–OH formation in OTS seems to weaken the strength of positive charges (–CH₃ groups) and thereby the dipoles between OTS and WSe₂ (also, p-doping phenomenon). However, after a 120 °C anneal, it is thought that the additional Si–O–Si bonds formed through the reaction between –OH groups strengthen the dipoles and eventually the p-doping phenomenon on WSe₂. For reference, the V_{TH} value shifted only slightly (around –0.2 V) after annealing at 120 °C in the control device (undoped), which was much smaller than the shifts (0.65–1.1 V) in doped devices. However, an encapsulation layer that can effectively block moisture may ensure the air-stability of the p-doping phenomenon.

Characterization of OTS-Doped WSe₂ Optoelectronic Devices.

To investigate the effect of the OTS-based controllable p-doping method on optoelectronic device performance, we performed a photocurrent measurement on the doped WSe₂ devices using a laser source with a

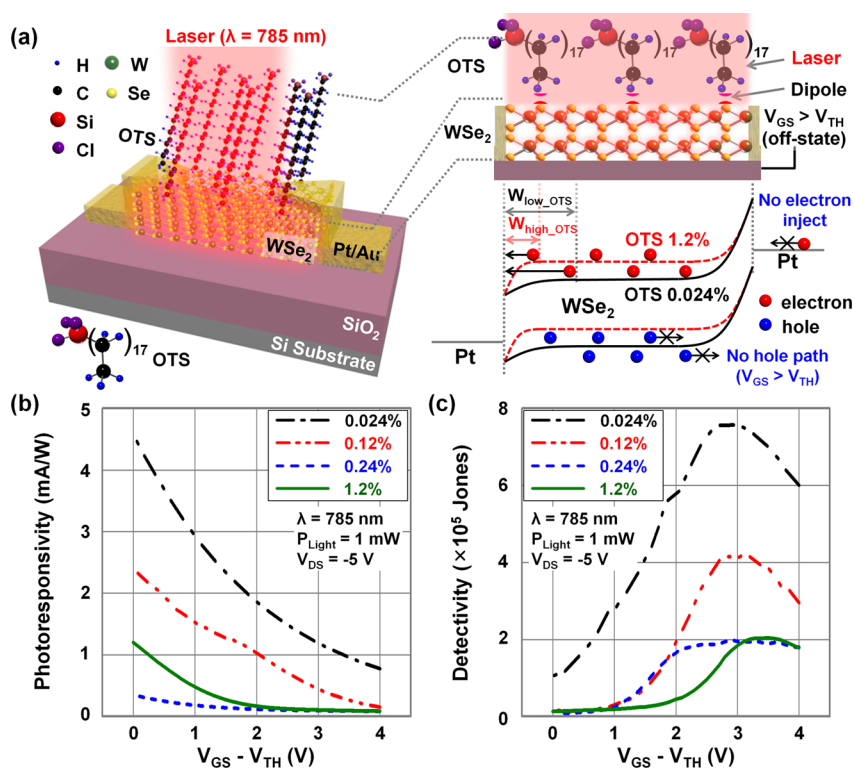


Figure 4. (a) Schematic diagrams showing the operation of an OTS-doped WSe₂ photodetector with a light source ($\lambda = 785$ nm and $P = 1$ mW) and the energy band diagrams of Pt–WSe₂–Pt junctions under the illuminated conditions. (b) Photoresponsivity and (c) detectivity as a function of $V_{GS} - V_{TH}$ (between 0 and 4 V in off-state) obtained on OTS-doped WSe₂ photodetectors ($V_{DS} = -5$ V).

wavelength of 785 nm. Figure 4a shows a schematic diagram of an OTS-doped WSe₂ photodetector device and the energy band diagram of Pt-undoped/doped WSe₂–Pt junctions with positive V_{GS} ($V_{GS} > V_{TH}$, off-state) and negative V_{DS} biases. As seen in the Supporting Information (Figure S3b), a photocurrent was not observed in the negative V_{GS} region (on-state) because of a high dark current level above 1.0×10^{-5} A/ μ m. In contrast, in the off-state (V_{GS} is positive and larger than V_{TH}), we clearly observed a photocurrent because the absence of hole paths in WSe₂ (low hole current) and the high electron barrier height (low electron current) reduced the dark current down to the order of 10^{-12} A/ μ m. When the level of p-doping was decreased by using a lower concentration OTS layer, a higher photocurrent was measured in the off-state. The increased depletion width seen on the left-side Pt–WSe₂ junction in Figure 4a might help collect more photogenerated electrons and eventually increase the photocurrent level in lightly doped WSe₂ devices ($W_{low-OTS} > W_{high-OTS} \rightarrow I_{Photo-low-OTS} > I_{Photo-high-OTS}$). The long diffusion length in lightly doped WSe₂ is also predicted to be another important reason for the increase in the photocurrent. We also note that the depletion width change in the right-side Pt–WSe₂ junction, which is relevant to hole carrier transport, can be neglected because the off-state WSe₂ region has no hole current path. Overall, the photocurrent increases as the p-doping level decreases because of

broader depletion width and increased diffusion length. This rising trend in photocurrent (as the p-doping level decreases) can also be confirmed in the photoresponsivity and detectivity analyses shown in parts b and c, respectively, of Figure 4. Because photoresponsivity ($R = I_{Photo}/P_{Light}$) is proportional to photocurrent, the highest R value (~ 4.5 mA/W) is observed in the 0.024% OTS-doped WSe₂ device. Photoresponsivity values decrease in all doped WSe₂ devices as $V_{GS} - V_{TH}$ increases up to 4 V because it becomes harder to collect electron carriers in the left-side Pt–WSe₂ junction when a higher $V_{GS} - V_{TH}$ bias is applied. The highest detectivity ($D^* = \sim 7.55 \times 10^5$ Jones) is observed in the WSe₂ device with the lowest doping concentration because the detectivity can be expressed as the ratio of I_{Photo} to I_{Dark} . Here, D^* is defined as $(RA^{1/2})/(2eI_{Dark})^{1/2}$, where R is photoresponsivity, A is the effective area of the device, e is the absolute value of the electron charge (1.6×10^{-19} C), and I_{Dark} is dark current. The decreased dark current at higher $V_{GS} - V_{TH}$ bias (Supporting Information, Figure S3) consequently increases the detectivity values in all of the doped WSe₂ devices. However, the detectivity starts decreasing beyond 3 V of $V_{GS} - V_{TH}$ because of an increase in dark current. The higher V_{GS} bias induces a higher electric field in the right-side Pt–WSe₂ junction, subsequently reducing the effective electron barrier height. In this photodetecting device experiment, we confirmed that it is possible to control the performance (photoresponsivity and detectivity) of

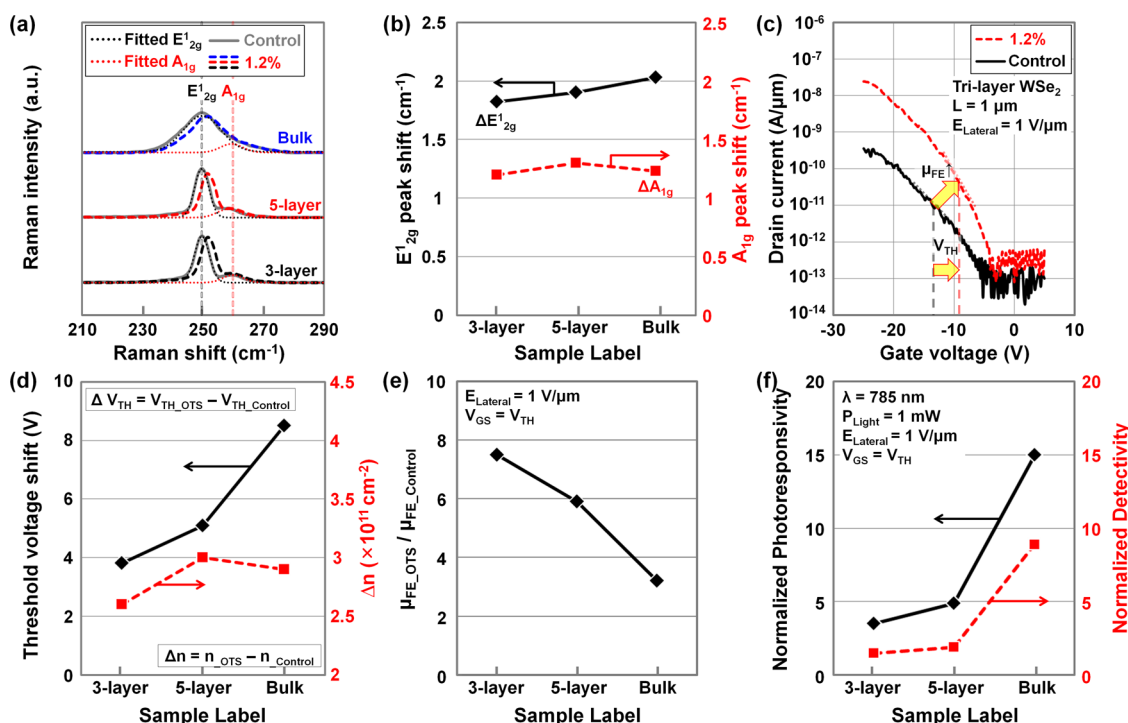


Figure 5. (a) Raman spectra of the 1.2% OTS-doped WSe₂ films and (b) extracted Raman peak shifts at E_{12g}^1 and A_{1g} obtained in trilayer, few layer, and bulk WSe₂ samples. (c) I_D – V_G characteristics of the transistors fabricated on (black) undoped and (red) 1.2% OTS-doped trilayer WSe₂ films ($L = 1 \mu\text{m}$ and $E_{\text{Lateral}} = 1 \text{ V}/\mu\text{m}$). Here, the I_D was normalized by channel width (W). (d) Threshold voltage shift ($\Delta V_{\text{TH}} = V_{\text{TH,OTS}} - V_{\text{TH,Control}}$) and carrier concentration increase ($\Delta n = n_{\text{OTS}} - n_{\text{Control}}$) of trilayer, few layer, and bulk WSe₂ device samples. (e) Field-effect mobility ratio ($\mu_{\text{FE,OTS}}/\mu_{\text{FE,Control}}$ at $V_{\text{GS}} = V_{\text{TH}}$) of trilayer, few layer, and bulk WSe₂ device samples. (f) Normalized photoresponsivity and detectivity of the 1.2% OTS-doped photodetectors fabricated on trilayer, few layer, and bulk WSe₂ films ($\lambda = 785 \text{ nm}$, $P_{\text{Light}} = 1 \text{ mW}$, $E_{\text{Lateral}} = 1 \text{ V}/\mu\text{m}$, and $V_{\text{GS}} = V_{\text{TH}}$).

an optoelectronic device (here, a photodetector) by adjusting the doping level of the TMD layer because the change in doping concentration affects the photocurrent *via* the depletion width (W) and the carrier diffusion length (L).

OTS Doping Characterization on WSe₂ Films of Various Thicknesses. To investigate the effect of WSe₂ thickness on OTS doping, we performed once again the Raman analysis and electrical/optical measurements on trilayer, few layer, and bulk WSe₂ device samples. As shown in Figure 5a, OTS doping was confirmed on the WSe₂ films of various thicknesses through Raman spectroscopy. Although a single peak was previously found at $\sim 250 \text{ cm}^{-1}$ on bulk WSe₂, one more peak was observed at 260 cm^{-1} on tri- and few layer WSe₂ films, which is thought to be the A_{1g} peak. For reference, Zeng *et al.* previously reported that two different WSe₂ Raman peaks appeared near 250 cm^{-1} (the E_{12g}^1 peak) and 260 cm^{-1} (the A_{1g} peak).²⁵ After fitting the E_{12g}^1 and A_{1g} Raman peaks (Supporting Information, Figure S6), we plotted ΔE_{12g}^1 and ΔA_{1g} values extracted from the Raman peaks before/after 1.2% OTS doping, as shown in Figure 5b. Regardless of the number of WSe₂ layers, almost the same degree of peak shift was confirmed (1.8 – 2.0 cm^{-1} for ΔE_{12g}^1 and 1.2 – 1.3 cm^{-1} for ΔA_{1g}), indicating that the OTS doping phenomenon is independent of the thickness of WSe₂. We then fabricated transistors on the three kinds of WSe₂ films

and performed electrical/optical measurements. As shown in Figure 5c, even in the trilayer WSe₂ device, a positive shift in threshold voltage and an increase in field-effect mobility were observed after p-doping using a 1.2% OTS layer. The ΔV_{TH} values showed a decreasing tendency as the number of layers reduced (Figure 5d). It is thought that the lower intrinsic carrier concentration (n_i) of trilayer WSe₂ with larger energy bandgap makes the Fermi-level (E_F) farther away from the intrinsic energy level (E_i) than it is in the other WSe₂ samples. Subsequently, a stronger electric field that enables an increase in tunneling probability from metal to WSe₂ is expected on the metal–WSe₂ junction at the equilibrium state. Therefore, the additional electric field induced by OTS doping seems to exert less influence on the trilayer device sample, eventually producing the lowest ΔV_{TH} value ($\sim 3.81 \text{ V}$). However, in the case of 2D sheet doping concentration, similar values between 2.62×10^{11} and $3.0 \times 10^{11} \text{ cm}^{-2}$ in Δn ($= n_{\text{OTS}} - n_{\text{Control}}$) were observed. Those values agree with the Raman-analyzed results because the number of carriers supplied by OTS doping is independent of the thickness of the WSe₂ films. The field-effect mobility values of the doped devices normalized by those of control devices are between 3.2 and 7.5, as shown in Figure 5e, and it consequently indicates that mobility was enhanced through OTS doping in all WSe₂ devices. Compared to a WSe₂ top-surface exposed to air

(control device), top-surface passivation by an OTS coat seems to suppress top-surface scattering and thereby enhance mobility. In particular, this mobility improvement is expected to be more effective in thinner devices in which mobility degradation through top-surface scattering is more severe: the $\mu_{\text{FE-OTS}}/\mu_{\text{FE-Control}}$ value increases from 3.2 to 7.5 as the WSe_2 layer becomes thinner. Figure 5f shows the photoresponsivity and detectivity of the three kinds of WSe_2 devices (at $V_{\text{GS}} = V_{\text{TH}}$) normalized by the values of the control devices (before OTS doping). As already analyzed in the bulk WSe_2 device (Figure 4), enhancement in photoresponsivity and detectivity was also observed in the tri- and few layer devices, although those showed a decreasing tendency as the number of WSe_2 layers declined (normalized R from 15.2 to 3.5 and normalized D^* from 8.9 to 1.5). This enhancement occurs because a photocurrent increases with a broader depletion width and an increased diffusion length in the metal- WSe_2 junction region, which are achieved by OTS doping. However, in the thinner WSe_2 devices, the reduction in the amount of absorbed light seems to decrease the photoresponsivity and detectivity.

CONCLUSIONS

In conclusion, we demonstrated a controllable nondegenerate p-doping technique for 2D TMD

materials (here, WSe_2) by adjusting the concentration of OTS in hexanes. This p-doping phenomenon is thought to originate from the fact that the methyl ($-\text{CH}_3$) functional groups in OTS have a positive pole and consequently reduce the electron carrier density in WSe_2 . The controlled p-doping levels were between 2.1×10^{11} and $5.2 \times 10^{11} \text{ cm}^{-2}$ in the nondegenerate regime in which WSe_2 -based electronic and optoelectronic devices can be properly designed. We also investigated the correlation between the controlled p-doping level and the performance parameters of WSe_2 -based electronic and optoelectronic devices ($V_{\text{TH}}\uparrow$, on-/off-currents $\uparrow$, field-effect mobility $\uparrow$, photoresponsivity $\uparrow$, and detectivity $\uparrow$ as the doping level increases). In addition, the p-doping effect by OTS was sustained in air for a long time with only small changes in the device performance (18–34% loss of ΔV_{TH} initially achieved by OTS doping for 60 h), and the degradation was almost completely recovered by additional thermal annealing at 120 °C. We also used Raman spectroscopy and electrical/optical measurements to confirm that OTS doping was independent of the thickness of the WSe_2 films. We expect our controllable p-doping method to make possible successful integration of future layered semiconductor devices.

EXPERIMENTAL METHODS

Formation and Doping Control of OTS on WSe_2 Film. Various thicknesses of WSe_2 nanoflakes (trilayer, few layer, and bulk) were transferred onto a 90 nm thick dry-oxidized SiO_2 layer grown on a heavily doped p-type Si substrate (resistivity $<0.005 \Omega\text{-m}$) by adhesive tape, and the samples were washed with acetone for 1 h to remove the tape residue. The number of layers for trilayer, few layer, and bulk WSe_2 flakes was identified by AFM (Supporting Information, Figure S5). For the formation of the OTS layer on WSe_2 samples, different amounts of OTS (12, 60, 120, and 600 μL) were added to 50 mL of hexane (a mixture of isomers), and the WSe_2 samples were soaked in each solution for 1 h. After that, the samples were rinsed with toluene, acetone, and deionized water several times and baked at 120 °C for 20 min.

Characterization of OTS-Doped WSe_2 Films. OTS/ WSe_2 / SiO_2 /Si samples were investigated and compared with a control sample (WSe_2 / SiO_2 /Si) by FT-IR (Bruker IFS-66/S) and PL/Raman spectroscopy (Alpha300 M+, WITec) measurements. The FT-IR spectral range was between 4000 and 20 cm^{-1} , the scan rate was 110 scans/s, and the resolution was above 0.1 cm^{-1} . A Raman spectroscope with an excitation wavelength of 532 nm was used, the laser beam size was 0.7–0.9 μm , and the instrumental spectral resolution was below 0.9 cm^{-1} . An integration time of 5 s and a spectrometer with 1800 grooves/mm were used for the test.

Fabrication and Electrical Characterization of an OTS-Doped WSe_2 Electronic Device. For the fabrication of back-gated WSe_2 transistors, we patterned source/drain electrodes (channel length and width of 5 μm) on WSe_2 / SiO_2 /Si samples by optical lithography, followed by Pt (10 nm) and Au (50 nm) deposition in an e-beam evaporator. To fabricate the transistor devices on thin (trilayer or few layer) WSe_2 films, e-beam lithography instead of optical lithography. Transistors were doped (or coated) by different concentrations of OTS (0.024%, 0.12%, 0.24%, and 1.2%) and were electrically analyzed using an HP 4415B semiconductor

parameter analyzer ($I_{\text{D}}-V_{\text{G}}$ and $I_{\text{D}}-V_{\text{D}}$) and an HP 4284A precision LCR meter (C-V). The threshold voltage (V_{TH}), carrier concentration (n), and field-effect mobility (μ_{FE}) were calculated from the $I_{\text{D}}-V_{\text{G}}$ data. Parameter extraction is explained in the Supporting Information (Figure S1). We used the equations $n = I_{\text{D}}/qW\mu V_{\text{DS}}$ and $\mu_{\text{FE}} = L/(WV_{\text{DS}}C_{\text{OX}}) \times (\partial I_{\text{D}}/\partial V_{\text{GS}})$ at $V_{\text{GS}} = V_{\text{TH}}$, where q is the electron charge, L and W are the length and width of the channel, respectively, and C_{OX} is $\epsilon_{\text{OX}} \times \epsilon_0/t_{\text{OX}}$, which is the gate oxide capacitance per unit area.

Characterization of an OTS-Doped WSe_2 Optoelectronic Device. To investigate the optoelectronic properties of the fabricated OTS-doped WSe_2 devices, a current–voltage ($I_{\text{D}}-V_{\text{G}}$) measurement was performed under dark and illuminated conditions. The light sources were set up using a dot laser with a wavelength of 785 nm and an optical power of 1 mW. For the characterization and comparison of the WSe_2 optoelectronic devices doped with OTS at different concentrations (0.024%, 0.12%, 0.24%, and 1.2%), photoresponsivity (R) and detectivity (D^*) were calculated from the $I_{\text{D}}-V_{\text{G}}$ curves: $R = I_{\text{photo}}/P_{\text{Light}}$ and $D^* = (RA^{1/2})/(2eI_{\text{Dark}})^{1/2}$ at the off-state ($V_{\text{GS}} > V_{\text{TH}}$), where I_{photo} is the generated photocurrent, P_{Light} is the total incident optical power, A is the effective area of detector, e is the absolute value of the electron charge ($1.6 \times 10^{-19} \text{ C}$), and I_{Dark} is the dark current.

Conflict of Interest: The authors declare no competing financial interest.

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Supporting Information Available: Extraction of electrical parameters (V_{TH} , μ_{FE} , n , and on-current). $C-V$ characterization of undoped and OTS-doped WSe_2 TFTs. Extraction of photoresponsivity (R) and detectivity (D^*). On-current monitoring of the OTS-doped WSe_2 TFTs. AFM analysis of the WSe_2 flakes used in this experiment. Fitting lines for the E_{12g}^1 and A_{1g} peaks of control and OTS 1.2%-doped WSe_2 samples. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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